

RUT Geometry-Light-Time Coupling Attempt v2

Stability, conservation, and testable consequences - technical upload note

Status

This v2 note pushes the v1 derivation scaffold one level deeper. It does not claim a finished theory. It asks whether the proposed geometry-light-time coupling can survive the first health checks a reviewer would expect: conservation bookkeeping, no-ghost stability, null/light-flow consistency, the GR/LambdaCDM limit, and at least one observable test chain.

Core v2 target: action -> mixed term -> effective kinetic shift -> stability condition -> observable wave/ripple consequence.

1. Starting action and bridge scalar

Use the same first-pass action family:

$$S = \int d^4x \sqrt{-g} \left[M_{Pl}^2 R/2 + L_m + L_\omega + L_L + \xi B(a) L(\omega L) + \lambda L(g_{mn} L^m L^n) \right]$$

with

$$L_\omega = -1/2 K(\omega)(\nabla_\mu \omega)^2 - V(\omega)$$

$$Q = L^\mu \nabla_\mu \omega$$

$$L(\omega L) = -1/2 Q^2$$

The null/light-flow condition remains:

$$g_{mn} L^m L^n = 0$$

2. Conservation bookkeeping

Because the full action is diffeomorphism-invariant when all fields are varied consistently, the total stress-energy should obey:

$$\nabla_\mu T^{\mu n} = 0$$

However, individual sectors do not need to be separately conserved. RUT should treat the mixed term as an exchange channel between the omega sector and the light-flow sector:

$$\nabla_\mu T^{\mu n}_\omega = +J_n$$

$$\nabla_\mu T^{\mu n}_L + \xi \nabla_\mu [B(a) T^{\mu n}(\omega L)] = -J_n$$

so that

$$\nabla_\mu [T^{\mu n}_\omega + T^{\mu n}_L + \xi B(a) T^{\mu n}(\omega L)] = 0$$

Reviewer-safe meaning: the coupling is allowed to transfer energy-momentum between phase-order and light-flow, but it must not create an unaccounted violation of total conservation.

3. No-ghost and kinetic stability

In the homogeneous FRW reduction, $Q = \dot{\omega}$, so the mixed term shifts the effective kinetic coefficient:

$$K_{\text{eff}}(a, \omega) = K(\omega) + \xi B(a)$$

The first no-ghost condition is therefore:

$$K_{\text{eff}}(a, \omega) > 0$$

If $0 \leq B(a) \leq 1$ and $K(\omega) = K_0 > 0$, then a simple sufficient prior is:

$$\xi > -K_0$$

A more conservative scan prior is:

$$K_0 \geq 1, \quad -0.5 K_0 \leq \xi \leq +1.0 K_0$$

This keeps the coupling perturbative and avoids the immediate pathology of negative kinetic energy.

4. Gradient and propagation sanity check

The same mixed term can change effective propagation along the L^μ direction. A minimal reviewer-safe requirement is that the coupling should not produce superluminal or unstable gradient behavior without a stated effective-field interpretation. A first conservative condition is:

$$0 \leq \xi B(a) / K_{\text{eff}}(a, \omega) \ll 1$$

This does not prove full causal safety, but it gives the first modeling rule: keep the mixed term subdominant until the full characteristic structure is derived.

5. FRW health check

With $\omega = \omega(t)$ and $L^\mu = (1, \hat{n}/a)$, the effective density and pressure are:

$$\rho_{\omega}^{\text{eff}} = \frac{1}{2} K_{\text{eff}} \dot{\omega}^2 + V(\omega)$$

$$p_{\omega}^{\text{eff}} = \frac{1}{2} K_{\text{eff}} \dot{\omega}^2 - V(\omega)$$

and the Friedmann relation becomes schematically:

$$H_{\text{RUT}}^2 = (8\pi G/3)[\rho_m + \rho_r + \rho_{\Lambda} + \rho_{\omega}^{\text{eff}} + \rho_L] - k/a^2$$

This makes the geometry-light-time term cosmologically legible: it contributes through the effective omega-sector energy budget and can therefore affect $H(z)$, r_d , and late-time bridge behavior.

6. Standard limit and kill switch

The required correspondence limit is:

$$B(a) \rightarrow 0 \text{ implies } K_{\text{eff}} \rightarrow K(\omega)$$

If ω is frozen, decoupled, or its density is negligible, then:

$$\rho_{\omega}^{\text{eff}} \rightarrow 0 \text{ and } H_{\text{RUT}}^2 \rightarrow H_{\Lambda\text{CDM}}^2$$

Kill switch: if any chosen ξ , $B(a)$, or ω evolution predicts local deviations already ruled out by solar-system gravity tests, gravitational-wave speed constraints, or stable cosmological background evolution, that parameterization must be rejected.

7. Observable consequence chain

The v_2 derivation produces a practical chain from equation to observable:

Step	RUT object	Observable pressure point
1	$Q = L^{\mu} \nabla_{\mu} \omega$	Light-flow sampling of phase-order
2	$K_{\text{eff}} = K + \xi B(a)$	Effective kinetic and pressure shift
3	$H_{\text{RUT}}(z)$	Expansion history and H_0 consistency
4	$r_{\text{d}}^{\text{RUT}}$	BAO/ripple horizon modification
5	$\Delta \phi_{\text{GW}}$	Gravitational-wave phase or siren tests

8. Gravitational-wave test expression

A first phenomenological wave-propagation signature can be written as:

$$\Delta \phi_{\text{GW}} \sim \int \xi B(a) Q \, d\lambda$$

This should be treated cautiously. It is not yet a waveform model. It is a target diagnostic: if gravitational waves sample the same phase-order field, then the accumulated phase, inferred distance, or siren-based H_0 measurement could constrain $\xi B(a)Q$.

Null test: if future gravitational-wave data tightly constrain $\Delta \phi_{\text{GW}} \rightarrow 0$ across relevant baselines, then $\xi B(a)Q$ must be small or absent in the wave-propagation regime.

9. Ripple-horizon test expression

The same kinetic and background shift can be linked to the RUT horizon program:

$$r_{\text{d}}^{\text{RUT}} = \int_{z_{\text{d}}}^{\infty} c_{\text{eff}}(z)/H_{\text{RUT}}(z) \, dz$$

A stable enlarged-horizon reading requires that the transport gain beats the expansion cost, consistent with the previous closed r_{d} model:

$$\dot{r}_{\text{d}} > \dot{r}_{\text{d}}$$

v_2 interpretation: the mixed geometry-light-time term is one possible source of the structured transport or background shift, but it must be tested against the same r_{d} and $H(z)$ constraints.

10. Boundary surfaces

The scalar Q gives a clean mathematical marker for transition surfaces:

$$Q = L^{\mu} \nabla_{\mu} \omega = 0$$

Candidate interpretation: light-flow becomes tangent to a constant-phase surface. In RUT language, this can represent a phase boundary, horizon-like transition, bounce interface, or sign-change layer. This is not proof of black hole or bounce physics yet, but it is a useful mathematical target.

11. What v_2 strengthens

v_2 improves v_1 by adding: total conservation bookkeeping, a no-ghost condition, a conservative coupling prior, a GR/ Λ CDM kill switch, and two observable test expressions. The strongest compact result is:

$$Q = L^{\mu} \nabla_{\mu} \omega$$

$$K_{\text{eff}} = K + \xi B(a)$$

$$K_{\text{eff}} > 0$$

$$\Delta \phi_{\text{GW}} \sim \int \xi B(a)Q \, d\lambda$$

$$r_d^{\text{RUT}} = \int c_{\text{eff}}/H_{\text{RUT}} dz$$

12. What remains unresolved

The largest unresolved issue is still the definition and dynamics of L^μ . v3 should choose one interpretation and derive its field equation: true null vector, radiation-flow current, averaged photon momentum direction, gravitational-wave propagation vector, or effective light-cone structure. The second unresolved issue is whether the mixed term's full metric variation remains stable and causal beyond the homogeneous approximation.

Bottom line

The v2 health check does not prove RUT, but it shows the geometry-light-time coupling can be organized into a legitimate EFT-style research target: define the coupling, impose stability, preserve the standard limit, and expose the term to wave-based and ripple-horizon tests.

Suggested filename

RUT_Geometry_Light_Time_Coupling_Attempt_v2_Stability_Conservation_Tests.pdf